

Hadron physics — Exam 2026

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Presentation

You are assigned one reaction of the following list

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|---|---|
| (a) $\pi^- p \rightarrow \eta' \pi^- p$ | (b) $J/\psi \rightarrow \gamma K_S K_S \eta'$ |
| (c) $B^+ \rightarrow K^+ \pi^+ \pi^- J/\psi$ | (d) $\Lambda_b \rightarrow J/\psi K^- p$ |
| (e) $\gamma p \rightarrow \rho \pi^- \Delta^{++}$ | (f) $D_s^+ \rightarrow K^- K^+ \pi^+$ |
| (g) $\gamma p \rightarrow \Xi^- K^+ K^+$ | (h) $K^- p \rightarrow \Sigma^+ \pi^- \pi^0$ |
| (i) $\gamma p \rightarrow p \pi^0 \pi^0$ | (j) $e^+ e^- \rightarrow \pi^+ D^0 D^{*-}$ |

for which you prepare a presentation with an in-depth analysis. Please prepare **no more than 10 slides** that fit within a ~ 15 -minute session. You may use any material, computational tool, or language model tool to support your preparation.

1. Reaction overview

- What are the initial and final state particles?
- What is the quark content of involved particles?
- Is this a decay or a scattering process?
- Which interaction is responsible?
- Draw a (Feynman) diagram representing this reaction.

2. Experimental signature

- What tracks, neutral clusters, or displaced vertices would be observed?
- Which particles require particle identification (PID)?
- What are the most significant background processes?
- Which detectors or experiments worldwide could study this reaction?

3. Kinematics

- What is the threshold energy for the process?
- What is the dimension of the phase space integral?
- How many independent kinematic variables determine the differential cross section/decay width for an unpolarized process?

4. Spectroscopic interpretation

- Specify all relevant quantum numbers (spin, parity, isospin, etc.)
- Are there intermediate resonances? In which particle combinations and partial waves do they appear?
- Are the resonances ground states, excitations, or possibly exotics?

5. Spin structure of the amplitude

- How many helicity or spin degrees of freedom are involved?
- How many independent amplitudes contribute to the total amplitude?
- Suggest a suitable parameterization of the angular part (e.g. Wigner D -matrix, spherical harmonics, etc.).
- Suggest a suitable parameterization of the dynamics part (e.g., Breit-Wigner, Flatté, K -matrix, etc.).